Execution Screenshots: End-to-End Recommendation Data Pipeline

Roll No: 2025AG05312 | AIML ZG529 Data Management for Machine Learning, Assignment 2

GitHub repository: https://github.com/jayatimishra/dm4ml-recsys-2025AG05312

Terminal screenshots were taken on my own machine while running each stage in order from an empty data folder, using the commands in docs/RUN_AND_SCREENSHOT.md. Browser screenshots show the Prefect and MLflow UIs and the HTML validation report.

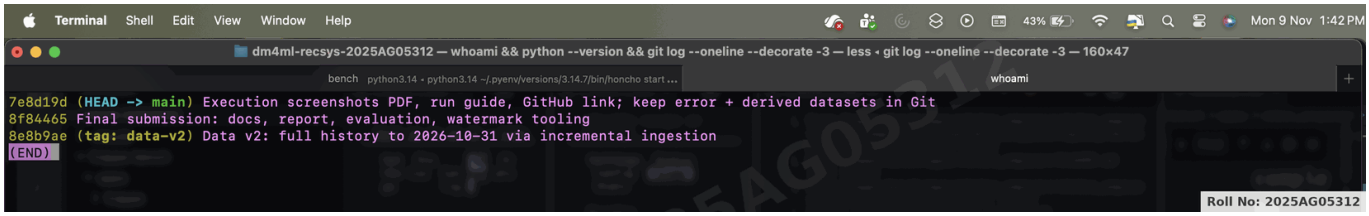
#	Stage	Command
0	Setup	`whoami && python --version && git log --oneline --decorate -3`
1	Source systems	`python -m src.ingestion.generate_sources --end-date 2026-11-01`
2	1 Ingestion	`python -m src.ingestion.ingest --mode incremental`
3	2 Validation	`python -m src.validation.validate`
4	3 Preparation	`python -m src.preparation.prepare`
5	4 Transformation	`python -m src.transformation.transform`
6	5 Feature store	`python -m src.features.feature_store`
7	6 Pre-training checks	`python -m src.validation.pre_training_checks`
8	7 Training	`python -m src.training.train`
9	8 Evaluation	`python -m src.evaluation.evaluate`
10	9 Serving	`python -m src.serving.serve --batch`
11	10 Orchestration	`python dags/run_pipeline.py | python dags/prefect_flow.py`
12	11 Versioning	`dvc dag && dvc status && git tag`

Every stage completed successfully; the full-DAG run in section 10 reports SUCCESS for all 10 tasks.

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Setup

Environment and Git history (tags data-v1, data-v2)

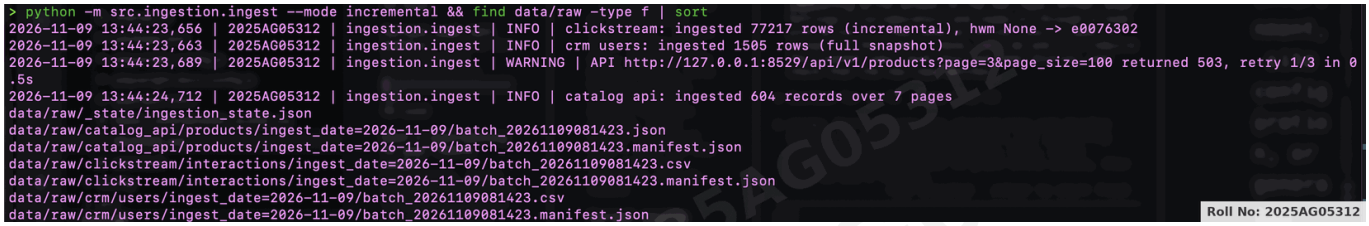


1. Data ingestion

Synthetic source systems generated (documented, seed 529)



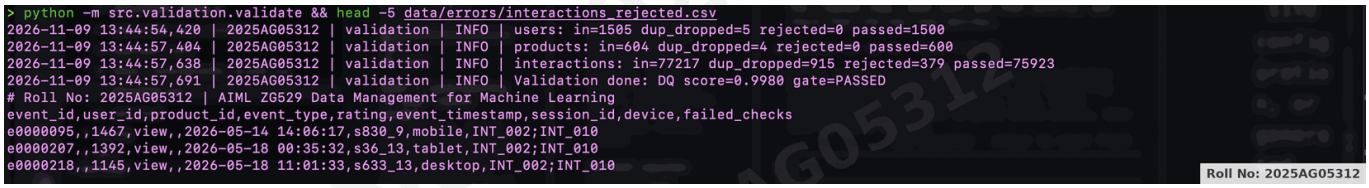
Ingestion of CSV exports + paginated REST API into the partitioned raw zone



The WARNING line is the deliberate HTTP 503 from the mock API, recovered by retry with backoff.

2. Data validation

Validation run, quality gate and the error dataset (data/errors)



Validation report (reports/validation/validation_report.html)

Data Validation Report

Roll No: 2025AG05312 | run at 2026-11-06T01:06:53.974335 | overall gate: **PASSED**

Entity summary

entity	rows_in	duplicates_dropped	rows_rejected	rows_passed
users	1505	5	0	1500
products	604	4	0	600
interactions	77217	915	379	75923

Quality gate (dataset-level)

entity	check	result	detail
users	schema_columns_present	PASS	missing=[] unexpected=[]
users	min_row_count	PASS	rows=1505 min=100
users	null_rate[user_id]<=0.0	PASS	null_rate=0.0000
users	critical_reject_rate<=5%	PASS	reject_rate=0.0000
products	schema_columns_present	PASS	missing=[] unexpected=[]
products	min_row_count	PASS	rows=604 min=100
products	null_rate[product_id]<=0.0	PASS	null_rate=0.0000
products	critical_reject_rate<=5%	PASS	reject_rate=0.0000
interactions	schema_columns_present	PASS	missing=[] unexpected=[]
interactions	min_row_count	PASS	rows=77217 min=1000
interactions	null_rate[user_id]<=0.01	PASS	null_rate=0.0020
interactions	null_rate[product_id]<=0.01	PASS	null_rate=0.0000
interactions	freshness<=7d	PASS	latest=2026-10-31 23:57:56 staleness_days=5
interactions	critical_reject_rate<=5%	PASS	reject_rate=0.0050

Row-level checks

entity	id	column	dimension	severity	description	checked	failed	pass rate
users	USR_UNQ	user_id	uniqueness	critical	user_id is unique (duplicates dropped, first kept)	1505	5	0.9967
users	USR_001	user_id	completeness	critical	user_id is not null	1500	0	1.0000
users	USR_002	signup_date	validity	critical	signup_date is a valid date	1500	0	1.0000
users	USR_003	age	completeness	warning	age is not null	1500	20	0.9867

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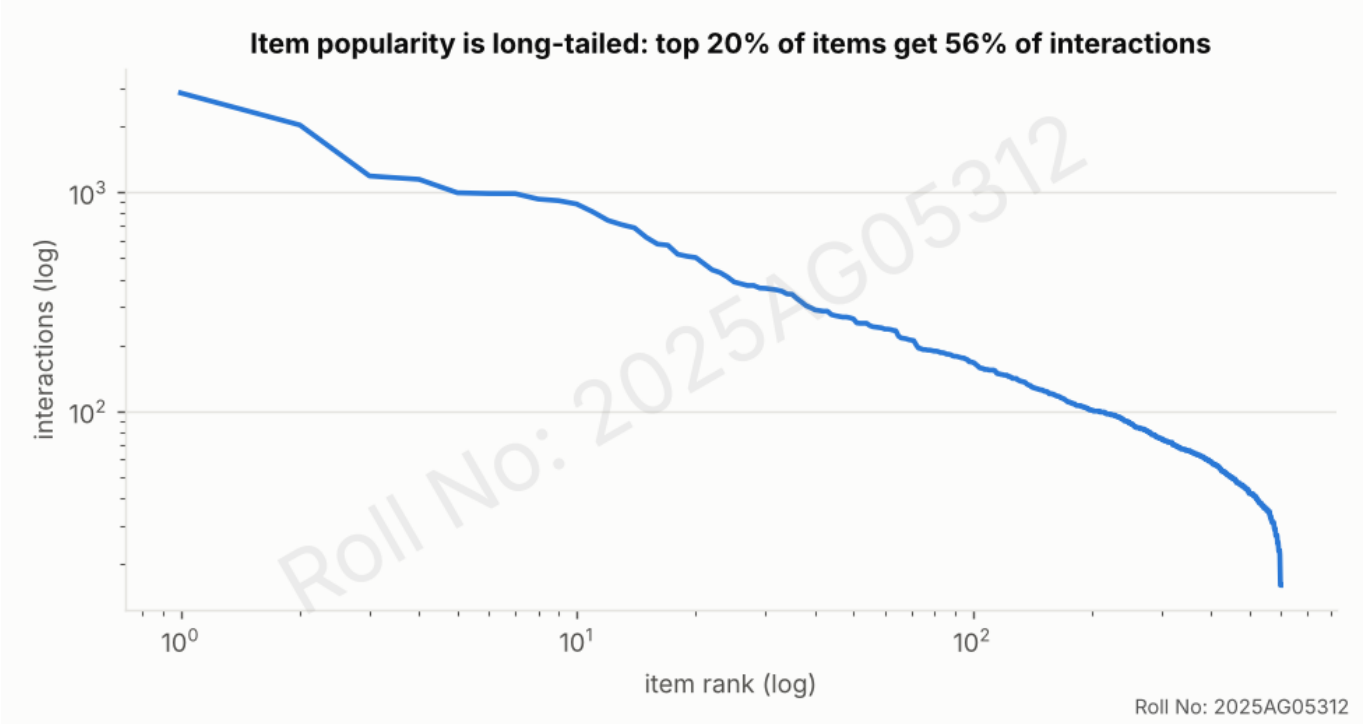
3. Data preparation

Cleaning, standardisation and EDA

```
> python -m src.preparation.prepare && ls reports/eda
2026-11-09 13:45:25,940 | 2025AG05312 | preparation | INFO | Prepared users=1500 products=600 interactions=75923; repairs={'users.age_imputed_with_median': 35,
'users.gender_set_unknown': 5, 'users.city_filled_unknown': 45, 'products.category_casing_fixed': 3, 'products.category_inferred_from_title': 12, 'products.price_text_parsed': 6, 'products.price_imputed_category_median': 15, 'products.price_winsorised_p99_5': 3, 'interactions.event_type_normalised': 305, 'interactions.rating_invalid_set_null': 53}
01_daily_events.png   03_item_long_tail.png 05_ratings.png        07_hour_of_day.png
02_funnel.png         04_user_activity.png  06_category_share.png eda_summary.json
```

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EDA: item long tail



4. Data transformation

Star schema load, point-in-time features, time-based splits, derived datasets

```
> python -m src.transformation.transform && ls data/processed/splits data/processed/derived
/Users/jayatimishra/Downloads/dm4ml-recsys-2025AG05312/src/transformation/transform.py:67: Pandas4Warning: DatetimeIndex.dayofweek is deprecated and will be rem
oved in a future version. Use DatetimeIndex.day_of_week instead.
  "month": dates.month, "day": dates.day, "day_of_week": dates.dayofweek,
/Users/jayatimishra/Downloads/dm4ml-recsys-2025AG05312/src/transformation/transform.py:68: Pandas4Warning: DatetimeIndex.dayofweek is deprecated and will be rem
oved in a future version. Use DatetimeIndex.day_of_week instead.
  "is_weekend": (dates.dayofweek >= 5).astype(int),
2026-11-09 13:45:58,424 | 2025AG05312 | transformation | INFO | Star schema loaded: {'dim_user': 1500, 'dim_product': 600, 'dim_date': 180, 'fact_interactions':
 75923}
2026-11-09 13:45:59,092 | 2025AG05312 | transformation | INFO | Features materialised as_of=2026-10-04 00:00:00: {'feat_user': 1500, 'feat_item': 600, 'agg_user
_item': 26433, 'feat_user_category': 9371}
2026-11-09 13:45:59,981 | 2025AG05312 | transformation | INFO | Features materialised as_of=2026-10-18 00:00:00: {'feat_user': 1500, 'feat_item': 600, 'agg_user
_item': 33353, 'feat_user_category': 10891}
2026-11-09 13:46:01,144 | 2025AG05312 | transformation | INFO | Features materialised as_of=2026-11-01 00:00:00: {'feat_user': 1500, 'feat_item': 600, 'agg_user
_item': 40220, 'feat_user_category': 11883}
2026-11-09 13:46:03,771 | 2025AG05312 | transformation | INFO | Splits exported: {'train': 61383, 'valid': 13680, 'test': 14540}
data/processed/derived:
item_popularity.parquet            matrix_user_index.parquet          user_item_interactions_long.parquet
matrix_item_index.parquet          user_activity_summary.parquet      user_item_matrix_train.npz

data/processed/splits:
test_interactions.parquet  train_interactions.parquet valid_interactions.parquet
```

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5. Feature management

Feature registry, offline/online materialisation and an online lookup

```
> python -m src.features.feature_store && head -20 reports/feature_catalog.md
2026-11-09 13:46:27,532 | 2025AG05312 | features | INFO | Feature store: 37 features in 4 views; {'user_profile_behaviour': {'valid': 1500, 'train': 1500, 'serv
e': 1500}, 'item_popularity_quality': {'valid': 600, 'train': 600, 'serve': 600}, 'user_item_interaction': {'valid': 26433, 'train': 33353, 'serve': 40220}, 'us
er_category_affinity': {'valid': 9371, 'train': 10891, 'serve': 11883}}
<!-- Roll No: 2025AG05312 -->
# Feature catalog

Generated from the feature registry (`data/features/registry.db`). All features are computed point-in-time in the warehouse (`src/transformation/sql/features.sq
l`) and served from the offline store for training and the online store for prediction.

## user_profile_behaviour (v1)

User demographics and engagement behaviour aggregated up to as_of. Entity key: `user_id`. Source table: `feat_user`. TTL: 30 days. Owner: ml-platform (2025AG053
12). Definition hash: `ff4af9216c56`.

| Feature | Type | Description | Used by |
|---|---|---|---|
| `u_n_events` | int | Total events by the user | serving enrichment / analysis |
| `u_n_views` | int | Product views | serving enrichment / analysis |
| `u_n_carts` | int | Add-to-cart events | serving enrichment / analysis |
| `u_n_purchases` | int | Purchases | serving enrichment / analysis |
| `u_conversion_rate` | float | Purchases / views | serving enrichment / analysis |
| `u_avg_rating_given` | float | Mean explicit rating the user gave | serving enrichment / analysis |
| `u_n_distinct_items` | int | Distinct products interacted with | serving enrichment / analysis |
| `u_active_days` | int | Distinct active days | serving enrichment / analysis |
| `u_days_since_last_event` | float | Recency in days at as_of | serving enrichment / analysis |
```

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6. Pre-training validation

Pandera schemas, temporal leakage check, PSI drift

```
> python -m src.validation.pre_training_checks
2026-11-09 13:47:12,516 | 2025AG05312 | validation.pre_training | INFO | pandera:train_interactions            PASS
2026-11-09 13:47:12,516 | 2025AG05312 | validation.pre_training | INFO | pandera:test_interactions             PASS
2026-11-09 13:47:12,516 | 2025AG05312 | validation.pre_training | INFO | pandera:user_features@train           PASS
2026-11-09 13:47:12,517 | 2025AG05312 | validation.pre_training | INFO | pandera:item_features@train           PASS
2026-11-09 13:47:12,517 | 2025AG05312 | validation.pre_training | INFO | pandera:user_item_features@train      PASS
2026-11-09 13:47:12,517 | 2025AG05312 | validation.pre_training | INFO | temporal_leakage                      PASS
2026-11-09 13:47:12,517 | 2025AG05312 | validation.pre_training | INFO | test_referential_integrity           PASS
2026-11-09 13:47:12,517 | 2025AG05312 | validation.pre_training | INFO | drift_psi:event_type_mix              PASS 0.0001
2026-11-09 13:47:12,517 | 2025AG05312 | validation.pre_training | INFO | drift_psi:category_mix                PASS 0.0025
```

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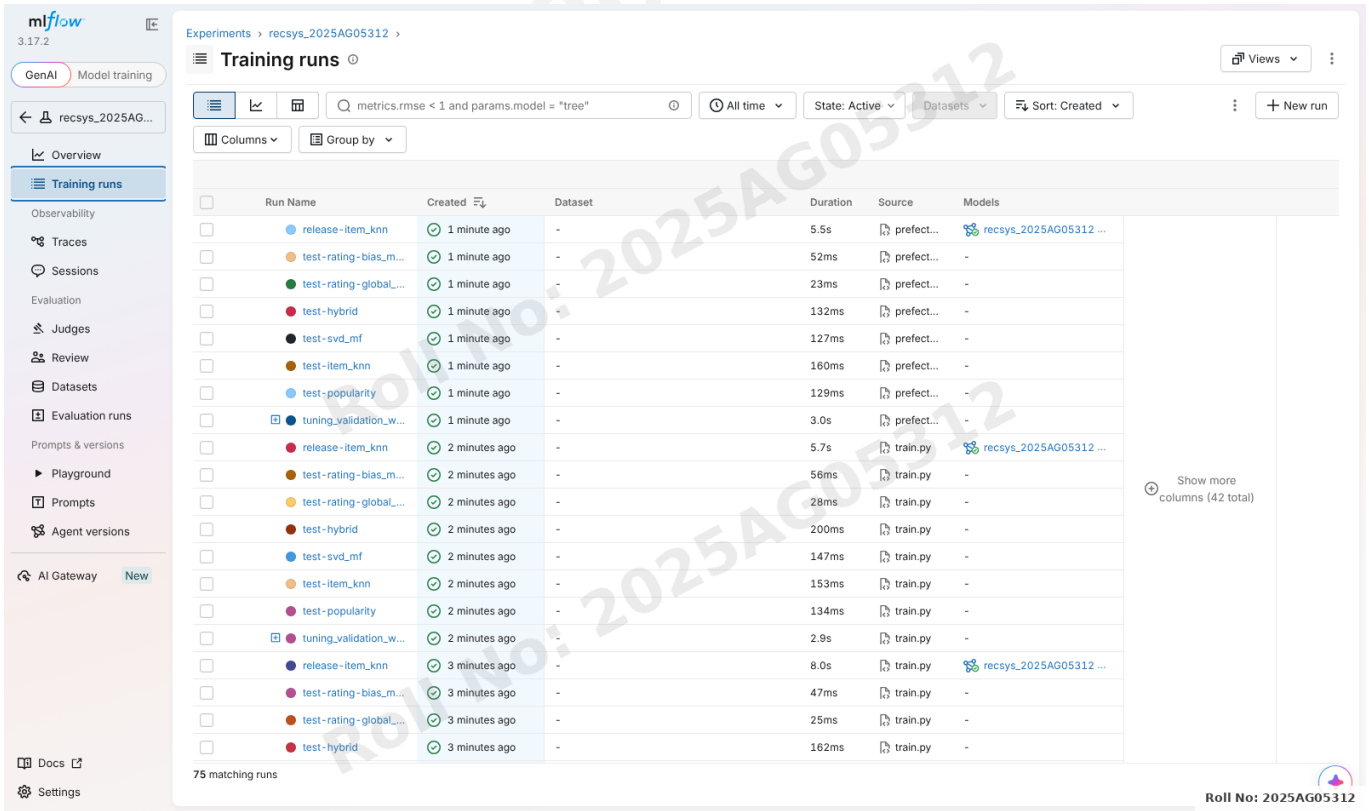
7. Model training and experiment tracking

Tuning on validation window, one-shot test, release + MLflow registry

```
> python -m src.training.train 2>&1 | tail -25
2026/11/09 13:48:23 INFO mlflow.store.db.utils: Updating database tables
2026-11-09 13:48:24,724 | 2025AG05312 | training | INFO | tune popularity {}                                    ndcg@10=0.0951
2026-11-09 13:48:24,848 | 2025AG05312 | training | INFO | tune item_knn   {'neighbours': 20}                   ndcg@10=0.1123
2026-11-09 13:48:24,963 | 2025AG05312 | training | INFO | tune item_knn   {'neighbours': 50}                   ndcg@10=0.1164
2026-11-09 13:48:25,078 | 2025AG05312 | training | INFO | tune item_knn   {'neighbours': 100}                  ndcg@10=0.1220
2026-11-09 13:48:25,199 | 2025AG05312 | training | INFO | tune item_knn   {'neighbours': 200}                  ndcg@10=0.1217
2026-11-09 13:48:26,081 | 2025AG05312 | training | INFO | tune item_knn   {'neighbours': 400}                  ndcg@10=0.1220
2026-11-09 13:48:26,233 | 2025AG05312 | training | INFO | tune svd_mf     {'factors': 16}                      ndcg@10=0.0974
2026-11-09 13:48:26,359 | 2025AG05312 | training | INFO | tune svd_mf     {'factors': 32}                      ndcg@10=0.0668
2026-11-09 13:48:26,481 | 2025AG05312 | training | INFO | tune svd_mf     {'factors': 64}                      ndcg@10=0.0467
2026-11-09 13:48:26,595 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 32, 'alpha': 0.1, 'beta': 0.05}  ndcg@10=0.0766
2026-11-09 13:48:26,708 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 32, 'alpha': 0.1, 'beta': 0.15}  ndcg@10=0.0908
2026-11-09 13:48:26,821 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 32, 'alpha': 0.25, 'beta': 0.05}  ndcg@10=0.0777
2026-11-09 13:48:26,933 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 32, 'alpha': 0.25, 'beta': 0.15}  ndcg@10=0.0917
2026-11-09 13:48:27,077 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 64, 'alpha': 0.1, 'beta': 0.05}  ndcg@10=0.0591
2026-11-09 13:48:27,210 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 64, 'alpha': 0.1, 'beta': 0.15}  ndcg@10=0.0795
2026-11-09 13:48:27,340 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 64, 'alpha': 0.25, 'beta': 0.05}  ndcg@10=0.0624
2026-11-09 13:48:27,464 | 2025AG05312 | training | INFO | tune hybrid     {'factors': 64, 'alpha': 0.25, 'beta': 0.15}  ndcg@10=0.0789
2026-11-09 13:48:28,114 | 2025AG05312 | training | INFO | Rating models: [{'model': 'global_mean', 'rmse': 0.9687889099967201, 'mae': 0.7922681998377661, 'n_rat
ings': 958}, {'model': 'bias_model', 'rmse': 0.9201453302164748, 'mae': 0.7587376392366119, 'n_ratings': 958}]
2026-11-09 13:48:28,115 | 2025AG05312 | training | INFO | Winner on test: item_knn {'neighbours': 100} ndcg@10=0.1244
2026/11/09 13:48:28 WARNING mlflow.pyfunc: Passing a Python object as `python_model` causes it to be serialized using CloudPickle, it requires exercising cautio
n as Python object serialization mechanisms may execute arbitrary code during deserialization.Consider using a file path (str or Path) instead. See https://mlfl
ow.org/docs/latest/ml/model/models-from-code/ for details.
2026/11/09 13:48:28 INFO mlflow.pyfunc: Inferring model signature from input example
Downloading artifacts: 100%|████████| 1/1 [00:00<00:00,  6.46it/s]
Successfully registered model 'recsys_2025AG05312'.
Created version '1' of model 'recsys_2025AG05312'.
```

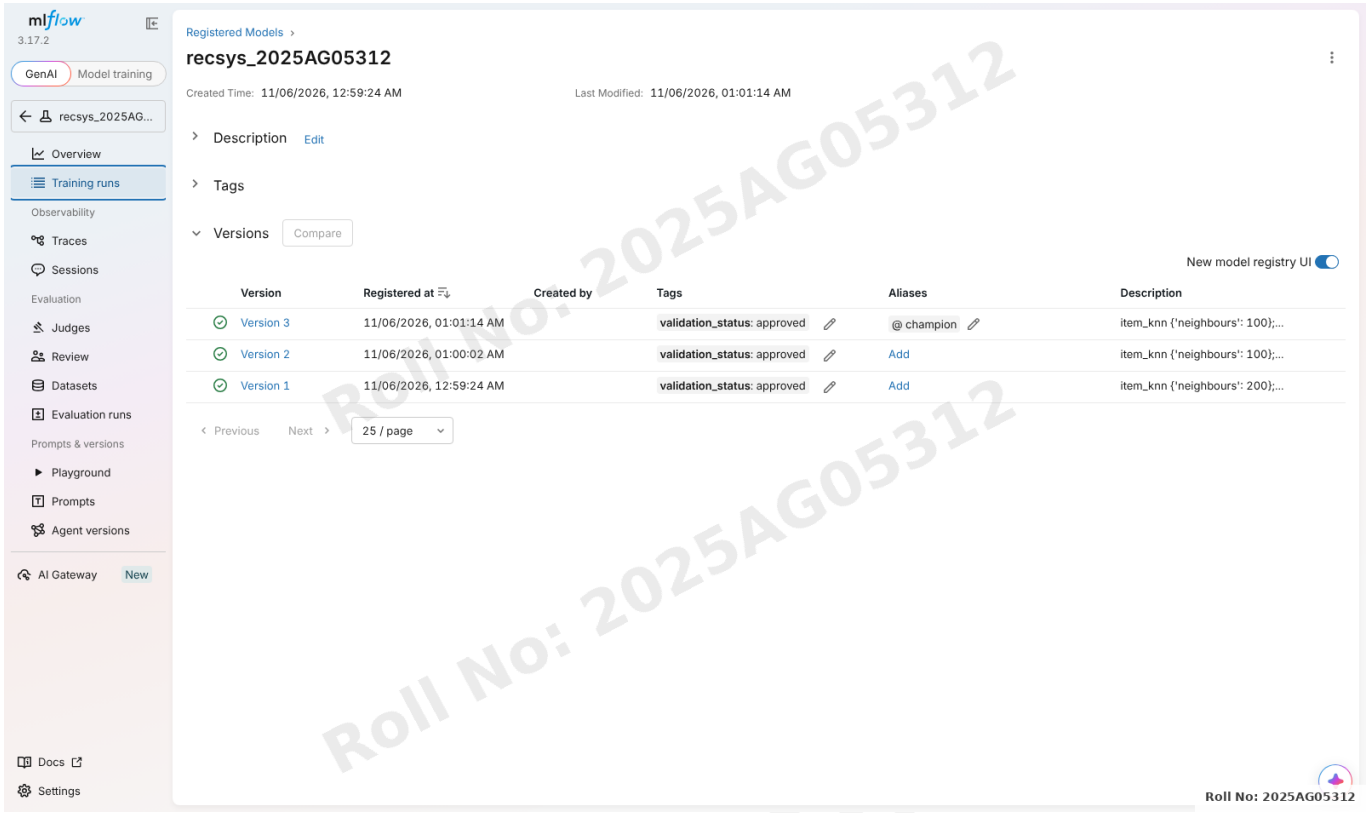
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MLflow: experiment runs



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MLflow: model registry with alias 'champion'



8. Model evaluation

Results table comparing all models

```
> python -m src.evaluation.evaluate && cat reports/model/results_table.md
2026-11-09 13:56:34,233 | 2025AG05312 | evaluation | INFO | Evaluation table written; winner=item_knn ndcg@10=0.1244 vs baseline 0.0851
<!-- Roll No: 2025AG05312 -->
# Model evaluation results (test window, K=10)

Evaluated on 921 users with at least one new add-to-cart or purchase in the 14-day test window. Selected model: **Item-KNN CF**.

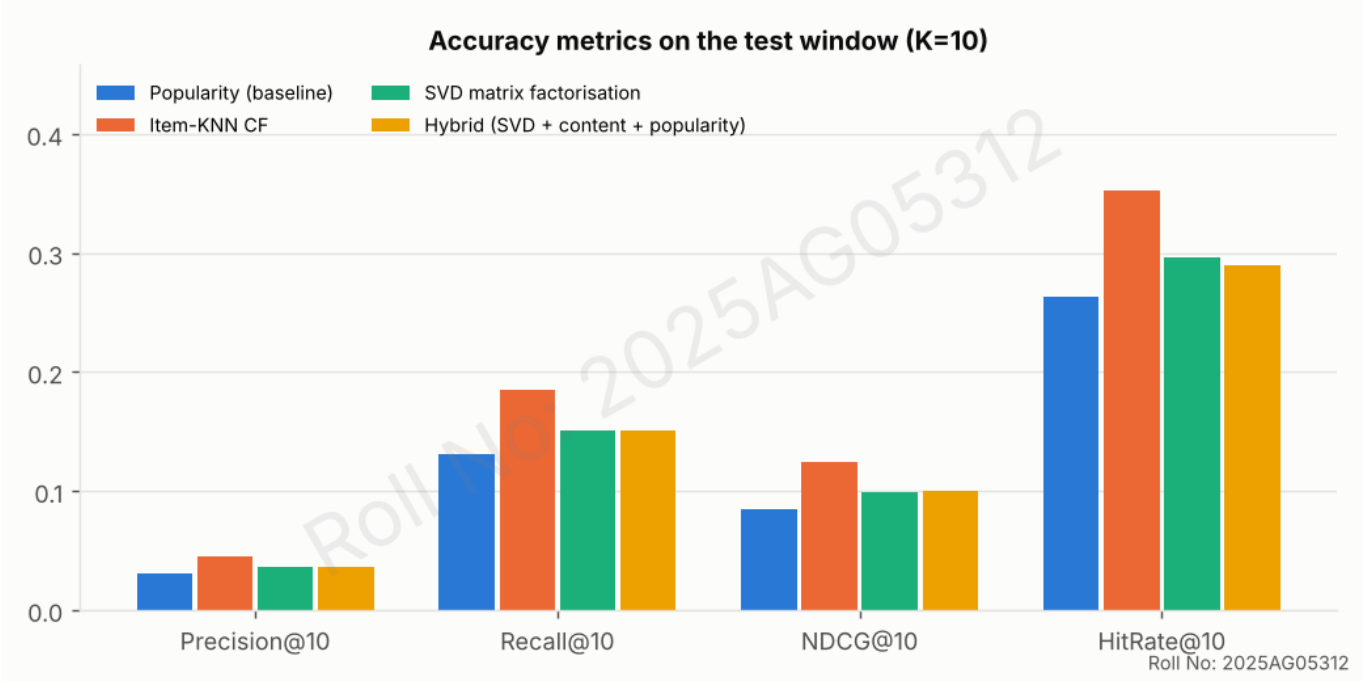
| Model | Params | precision@10 | recall@10 | ndcg@10 | map@10 | hit_rate@10 | coverage | diversity | novelty | NDCG lift |
|---|---|---|---|---|---|---|---|---|---|---|
| Popularity (baseline) | `{}` | 0.0312 | 0.1312 | 0.0851 | 0.0526 | 0.2628 | 0.0683 | 0.8963 | 6.69 | +0.0% |
| **Item-KNN CF** | `{"neighbours": 100}` | 0.0451 | 0.1850 | 0.1244 | 0.0789 | 0.3529 | 0.4550 | 0.7191 | 7.53 | +46.1% |
| SVD matrix factorisation | `{"factors": 16}` | 0.0367 | 0.1504 | 0.0990 | 0.0605 | 0.2964 | 0.3817 | 0.6526 | 8.01 | +16.3% |
| Hybrid (SVD + content + popularity) | `{"factors": 32, "alpha": 0.25, "beta": 0.15}` | 0.0359 | 0.1505 | 0.1007 | 0.0624 | 0.2899 | 0.7100 | 0.3407 | 8.46 | +
18.3% |

## Explicit rating prediction (supplementary)

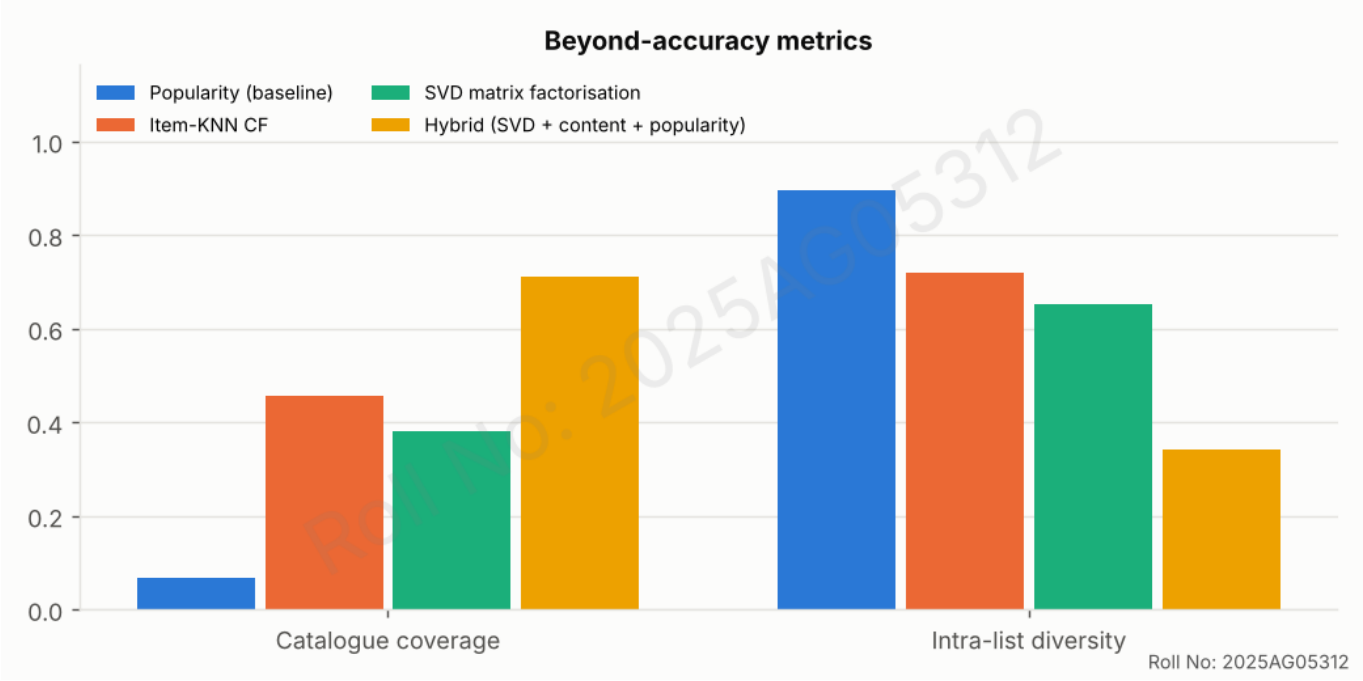
| Model | RMSE | MAE | Test ratings |
|---|---|---|---|
| global_mean | 0.9688 | 0.7923 | 958 |
| bias_model | 0.9201 | 0.7587 | 958 |
```

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Accuracy metrics



Beyond-accuracy metrics



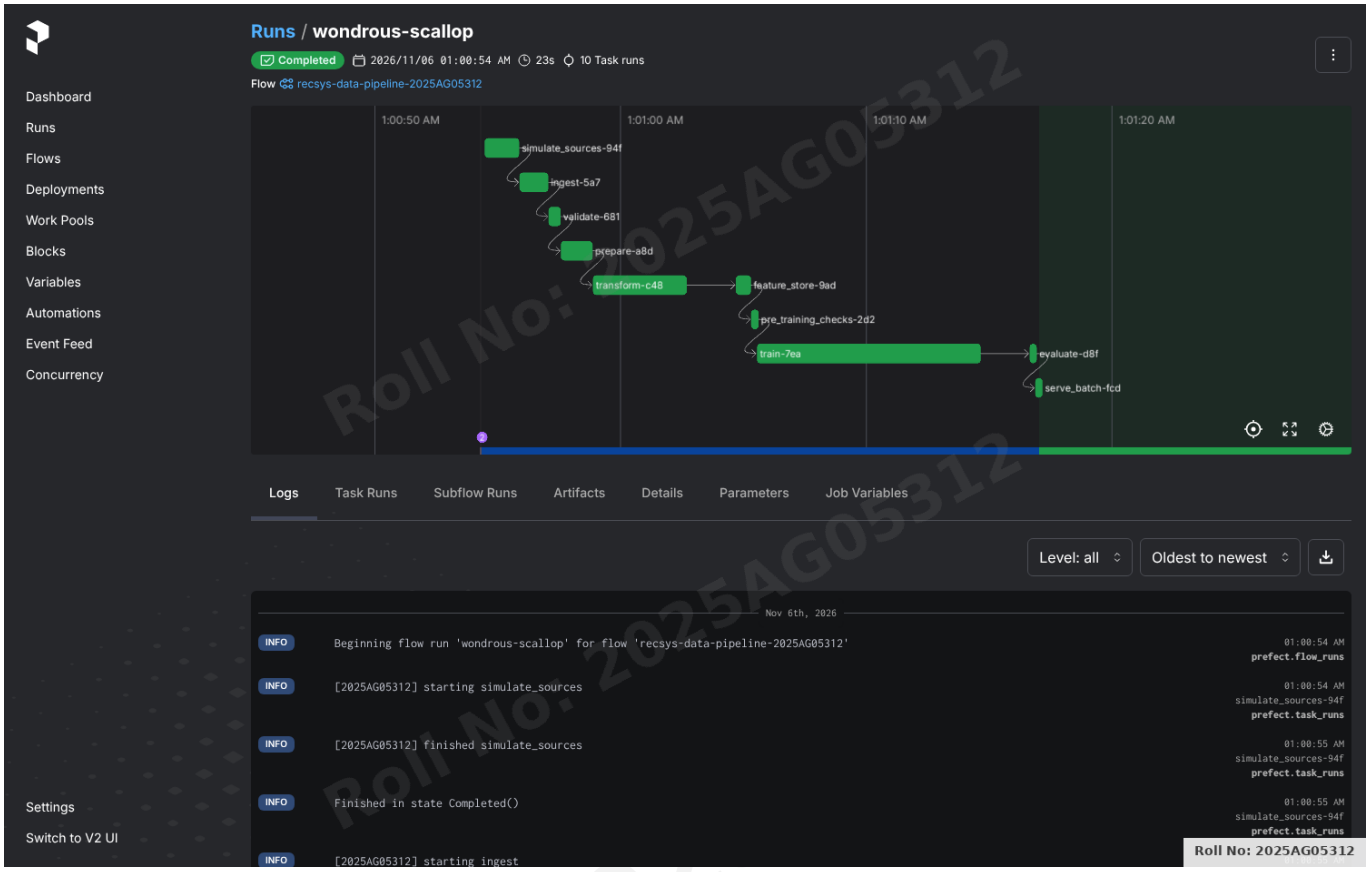
9. Prediction service

Batch recommendations and HTTP /recommend endpoint

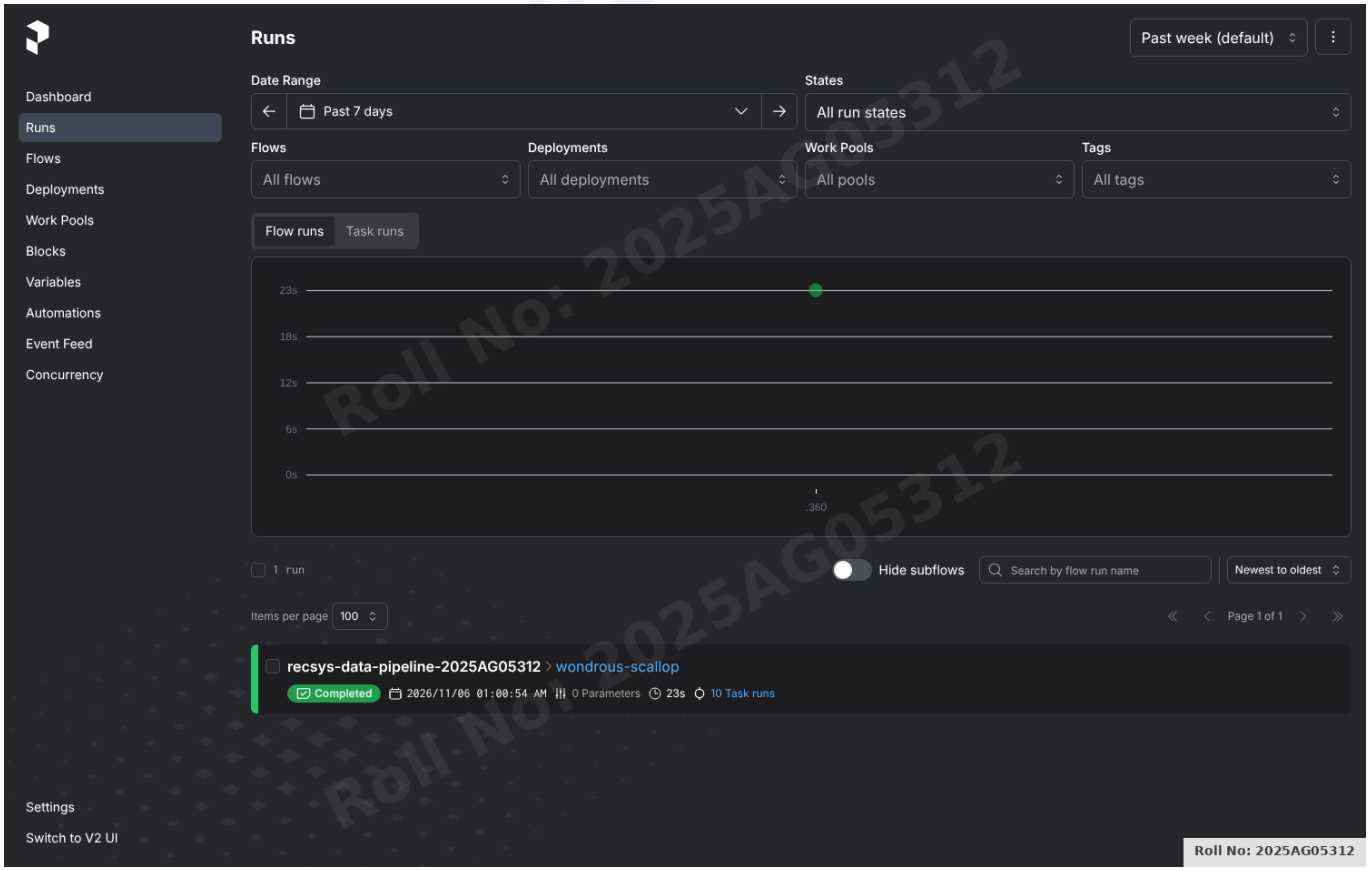
```
> python -m src.serving.serve --batch && head -12 reports/sample_recommendations.csv
/Users/jayatimishra/Downloads/dm4ml-recsys-2025AG05312/.venv/lib/python3.15/site-packages/mlflow/pyfunc/utils/data_validation.py:187: UserWarning: Add type hints to the `predict` method to enable data validation and automatic signature inference during model logging. Check https://mlflow.org/docs/latest/model/python_model.html#type-hint-usage-in-pythonmodel for more details.
  color_warning(
2026-11-09 13:57:10,979 | 2025AG05312 | serving | INFO | Batch recommendations for [7, 42, 314, 1001, 99999] written (model v1); user 99999 is a cold-start test
# Roll No: 2025AG05312 | AIML ZG529 Data Management for Machine Learning
user_id,rank,product_id,score,category,brand,price,i_avg_rating,in_stock,model_version
7,1,1510,1.05811,Automotive,BrandF,1479.35,3.558,1,1
7,2,1221,0.99098,Sports,BrandJ,2089.78,3.353,1,1
7,3,1211,0.98021,Music,BrandB,732.84,3.276,1,1
7,4,1376,0.96425,Automotive,BrandR,1517.91,3.788,1,1
7,5,1017,0.95119,Beauty,BrandP,589.74,4.15,1,1
7,6,1160,0.9403,Sports,BrandM,1907.79,3.277,1,1
7,7,1208,0.86735,Beauty,BrandT,829.31,3.619,1,1
7,8,1128,0.83092,Beauty,BrandM,348.37,4.264,1,1
7,9,1489,0.8228,Sports,BrandF,2602.11,3.302,1,1
7,10,1469,0.72584,Toys,BrandQ,1314.38,2.8,1,1
```

10. Orchestration and reproducibility

Prefect flow run: all tasks Completed



Prefect: run history



Full DAG run with the dependency-free runner

```
> python dags/run_pipeline.py 2>&1 | grep orchestrator
2026-11-09 13:57:47,384 | 2025AG05312 | orchestrator | INFO | task simulate_sources SUCCESS in 1.1s
2026-11-09 13:57:48,564 | 2025AG05312 | orchestrator | INFO | task ingest         SUCCESS in 1.2s
2026-11-09 13:57:48,999 | 2025AG05312 | orchestrator | INFO | task validate       SUCCESS in 0.4s
2026-11-09 13:57:50,465 | 2025AG05312 | orchestrator | INFO | task prepare        SUCCESS in 1.5s
2026-11-09 13:57:54,986 | 2025AG05312 | orchestrator | INFO | task transform      SUCCESS in 4.5s
2026-11-09 13:57:55,570 | 2025AG05312 | orchestrator | INFO | task feature_store  SUCCESS in 0.6s
2026-11-09 13:57:55,815 | 2025AG05312 | orchestrator | INFO | task pre_training_checks SUCCESS in 0.2s
2026-11-09 13:58:04,692 | 2025AG05312 | orchestrator | INFO | task train          SUCCESS in 8.9s
2026-11-09 13:58:04,916 | 2025AG05312 | orchestrator | INFO | task evaluate       SUCCESS in 0.2s
2026-11-09 13:58:04,995 | 2025AG05312 | orchestrator | INFO | task serve_batch    SUCCESS in 0.1s
2026-11-09 13:58:04,996 | 2025AG05312 | orchestrator | INFO | Pipeline run 20261109082746: SUCCESS
```

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DVC pipeline DAG and data-version diff

